

3. Water quality can be monitored using invertebrates as indicators.

(a) A method for an investigation to monitor water pollution using invertebrates is described below, however the steps are in the wrong order.

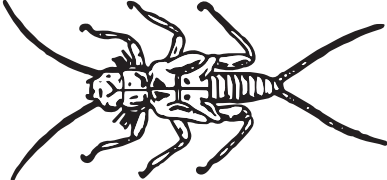
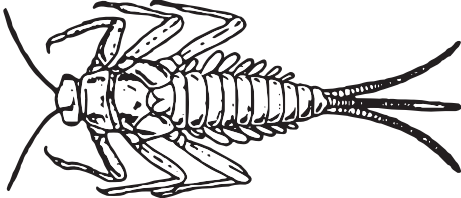
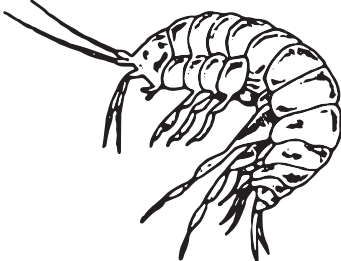
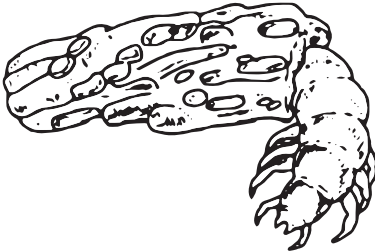
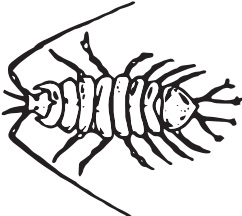
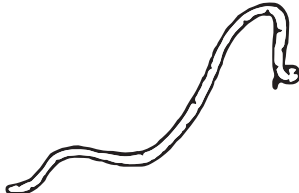
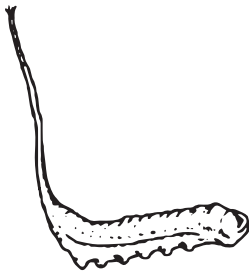
- A. Identify **two** locations to be tested e.g. two different areas in the same stream.
- B. Record the total biomass of each species of invertebrate that has been caught.
- C. Study the organisms in the tray and try to identify the invertebrates.
- D. Collect some water in a tray so it is about 2 to 3 cm deep.
- E. Collect samples of invertebrates using a net and transfer them to the tray.
- F. Pour contents gently back into the stream.

Arrange the steps in order by placing the letters **B**, **C**, **D** and **E** in the correct boxes below.

[3]

A					F
----------	-------	-------	-------	-------	----------

(b) Some invertebrate species used as indicators are shown below.

Water quality	Indicator species	
Clean water	 Stonefly nymph	 Mayfly nymph
Some pollution	 Freshwater shrimp	 Caddisfly larva
Moderate pollution	 Water louse	 Bloodworm
High pollution	 Sludgeworm	 Rat-tailed maggot
Very high pollution	No life	

(Not drawn to scale.)

The results from such an investigation are shown in the table below.

	Total biomass in sample (g)	
	Stream A	Stream B
Mayfly nymph	2	0
Caddisfly larva	55	0
Freshwater shrimp	78	2
Water louse	8	6
Bloodworm	6	22
Sludgeworm	1	110

Use the information in both tables to answer the following questions.

(i) I. Name the **two** most common invertebrates found in stream **A**. [1]

..... and

II. State the water quality in stream **A**. [1]

.....

(ii) Explain whether stream **B** has the same water quality as stream **A**. [2]

.....
.....
.....

(iii) State how the results would be different for a stream with very high pollution. [1]

.....